

Course Code: CSE 2203 (Database Management Systems), Marks: 20, Time: 50 Minutes

Scenario: Library Management System

A library needs a system to manage its operations, including book lending, member records, and staff assignments. The system requirements are as follows:

1. The library contains many **Books**. Each book has attributes like Book_ID, Title, Author, Publisher, and Year_Published.
2. **Members** can borrow books from the library. Each member has a Member_ID, Name, Address, Phone, and Membership_Date.
3. Each book can be borrowed by one member at a time, but it can be borrowed multiple times over its lifecycle. This relationship involves attributes like Borrow_Date and Return_Date.
4. The library employs **Staff** who manage the book lending process. Staff members have Staff_ID, Name, Position, and Hire_Date.
5. **Categories** organize books. Each category has a Category_ID and Category_Name, and a book can belong to one or more categories.

Answer the following question for the above scenario:

1. Identify entities, relationships, and attributes
2. Draw an E-R diagram